Electrochemistry Primer for ORR

with just enough physical chemistry to read the codebase

Onboarding doc 01 of 06

May 12, 2026

What this document is for

The longest of the onboarding docs. It builds, from first principles, the chemistry vocabulary you need to talk about the oxygen reduction reaction (ORR), the electric double layer, and the Butler–Volmer (BV) kinetic equation. Read it once now; you will return to specific sections as they come up. There are exercises at the end of each major section — do them in your notebook before the meeting; we will go over them together.

Contents

1 What an electrochemical cell is

An **electrochemical cell** is two pieces of metal (or other electronic conductor) called **electrodes** dipped into a liquid called the **electrolyte**. The electrolyte is a solvent (water for us) containing dissolved **ions**: charged particles that can move through the liquid. The two electrodes are connected externally through a wire and possibly a power supply.

Why is this interesting? Because *electrons can only flow in the wire, not in the electrolyte.* So whenever current goes around the loop, something has to convert electron flow at the electrode–electrolyte interface into ion flow in the electrolyte. That "something" is a chemical reaction at the interface. Different reactions happen at different rates and at different applied voltages, and that is the entire content of electrochemistry.

1.1 Half-cells, oxidation, reduction

Each electrode hosts a **half-reaction**. By convention, **reduction** (gain of electrons) happens at the **cathode**; **oxidation** (loss of electrons) happens at the **anode**. The classical mnemonic is "RED CAT, AN OX." For the ORR cell we model:

- **Cathode (the disk):** oxygen is reduced. $O_2 + nH^+ + ne^- \to$ products.
- **Anode (somewhere far away in the cell, often a Pt counter electrode):** water is oxidized to O_2, the so-called oxygen evolution reaction.

We model only the cathode region because that is where the interesting physics is. The counter-electrode is treated as a boundary condition implicitly through the bulk concentrations.

1.2 Faraday's law (the dimensional anchor)

The central conversion factor between electricity and chemistry is **Faraday's constant**:

$$F \approx 96\,485\ \ \mathrm{C/mol\ e^-}.$$

Faraday's law of electrolysis states that the amount of substance reacted is proportional to the charge passed:

$$\text{(moles reacted)} = \frac{Q}{n_e\, F},$$

where Q is the integrated current (in coulombs) and n_e is the number of electrons transferred per molecule. Differentiating:

$$i = n_e\, F\, r_{\text{rxn}}, \tag{1}$$

where i is the current density (A/m^2) and r_{rxn} is the molar reaction rate per unit area (mol/m^2/s). This equation is the bridge between the chemical models in our code and the *measured* quantity i.

1.3 Convention: cathodic current is negative

Electrochemistry sign conventions are a constant source of bugs. The convention this codebase uses (and the one most ORR papers use):

Sign convention

- Current density $i > 0$ for **anodic** (oxidative) flow.
- $i < 0$ for **cathodic** (reductive) flow — ORR makes $i < 0$.
- The disk-current curves you will see plotted are typically plotted as $|i|$ vs. V, so visually the cathodic "ramp up" goes *up*, even though the underlying i is going more negative.

2 Equilibrium electrochemistry

2.1 Standard reduction potential E°

Each half-reaction has a thermodynamic equilibrium voltage. By convention, half-reactions are written as reductions and tabulated relative to the **standard hydrogen electrode (SHE)**, which is defined to have $E^\circ \equiv 0$. For the ORR we care about two:

$$\mathrm{O_2} + 4\mathrm{H^+} + 4e^- \rightleftharpoons 2\,\mathrm{H_2O} \qquad E^\circ_{4e} = +1.23\ V_{\text{RHE}} \tag{2}$$

$$\mathrm{O_2} + 2\mathrm{H^+} + 2e^- \rightleftharpoons \mathrm{H_2O_2} \qquad E^\circ_{2e} = +0.695\ V_{\text{RHE}} \tag{3}$$

These are the values from Ruggiero 2022 §1, reported on the *reversible hydrogen electrode* (RHE) scale — the convention used throughout this codebase, the deck, and the ORR literature. The RHE scale absorbs the pH dependence of any proton-coupled electron transfer (PCET) reaction. Numerically, at pH $= 0$ these coincide with the standard reduction potentials vs SHE; at other pH the SHE-referenced values shift by $-0.059\,\mathrm{V} \cdot \mathrm{pH}$, exactly cancelling the RHE-vs-SHE shift, so the values quoted above are pH-independent. The next subsection makes the SHE/RHE conversion explicit.

The 4-electron pathway is more thermodynamically favorable (higher E°) but also requires breaking the strong O–O bond, which can be slow — hence the kinetic "why does the 2e pathway happen at all?" question that drives the field.

2.2 The Nernst equation

If the reactants and products are not at standard concentration (1 mol/L for solutes, 1 atm for gases), the equilibrium voltage shifts. For a generic half-reaction $\nu_O\,\mathrm{O} + n_e e^- \rightleftharpoons \nu_R\,\mathrm{R}$, the **Nernst equation** gives:

$$E_{\text{eq}} = E^\circ - \frac{RT}{n_e F}\ln\left(\frac{a_R^{\nu_R}}{a_O^{\nu_O}}\right), \tag{4}$$

where a_i is the **activity** of species i (close to concentration in dilute solution, with corrections we will discuss). The prefactor RT/F at room temperature is the **thermal voltage**:

$$V_T \;\coloneqq\; \frac{RT}{F} \approx 25.7\text{ mV} \quad \text{at } T = 298\,\text{K}.$$

You will see V_T everywhere in the code (`V_T` in `scripts/_bv_common.py`).

2.3 Reference electrodes: SHE, Ag/AgCl, RHE

In the lab, nobody uses a hydrogen electrode — it requires bubbling explosive H_2 at exactly 1 atm over a Pt black surface. Real experiments use practical references like **Ag/AgCl in saturated KCl**, then *convert* to a common pH-aware reference, the **reversible hydrogen electrode (RHE)**:

$$V_{\text{RHE}} = V_{\text{Ag/AgCl}} + 0.197\,\text{V} + 0.059\,\text{V}\cdot\text{pH}.$$

The 0.059 V/pH conversion is just $V_T \ln 10 \approx 59$ mV at room temperature, applied to the Nernstian pH dependence of the H_2 half-reaction. **Every voltage in our code and in the deck data is reported as V_{RHE}.** If you ever see a paper quoting voltages "vs. SHE" or "vs. Ag/AgCl," convert before comparing.

2.4 Overpotential η

When the applied voltage V at an electrode differs from the local equilibrium E_{eq}, a current flows. The driving force is the **overpotential**:

$$\eta \;\coloneqq\; V - E_{\text{eq}}. \tag{5}$$

Note the sign: $\eta < 0$ drives the reduction direction (cathodic), $\eta > 0$ drives oxidation (anodic). Most of the ORR voltage range we care about has $\eta < 0$.

Exercise 2.1. For the 4e ORR, the standard reduction potential at pH 0 is $E^\circ(\text{SHE}) = +1.229$ V. Apply the Nernst equation at bulk pH 4 ($[H^+] = 10^{-4}$ M, $[O_2] = 1.2 \times 10^{-3}$ M, $P_{O_2} = 1$ atm; activity of liquid water $= 1$) to compute E_{eq} vs. SHE. Then convert to V_{RHE} using the RHE/SHE relation in §2.3. You should recover $E_{\text{eq}} \approx +1.23\ V_{\text{RHE}}$, confirming that the RHE-referenced potential of any PCET reaction with $n_{H^+}/n_e = 1$ is pH-independent.

3 The electric double layer (EDL)

This section is the bridge between equilibrium thermodynamics and the much stranger world of the electrode-electrolyte *interface*.

3.1 Why an interface develops a charge layer

When a metal electrode is immersed in an electrolyte and held at some applied potential V, electrons accumulate (or deplete) at the metal surface. To maintain overall electroneutrality on the solution side within a few nanometers, ions of the opposite sign crowd toward the surface and ions of the same sign are repelled. The result is a thin region of net charge in the electrolyte adjacent to the metal, balancing the surface electronic charge: this is the **electric double layer (EDL)**.

The EDL controls almost everything we care about:

- It sets the local concentrations of H^+, cations, etc. at the electrode surface — where the chemistry happens.
- It absorbs a portion of the applied voltage, so the "available driving force" for the reaction is less than $V - E_{\mathrm{eq}}^{\mathrm{bulk}}$.
- It depends on cation identity and salt concentration, which is one mechanism for the cation effect.

3.2 Three increasingly realistic EDL models

Helmholtz (1853). A single rigid sheet of counter-ions sits one ionic radius from the metal, like a parallel-plate capacitor. Predicts a constant capacitance independent of voltage and concentration. Wrong, but simple.

Gouy–Chapman (1910–13). The counter-ion "sheet" is allowed to spread out as a continuous **diffuse layer** obeying a Boltzmann distribution:

$$c_i(x) = c_i^{\mathrm{bulk}} \exp\left(-\frac{z_i F \phi(x)}{RT}\right), \tag{6}$$

where $\phi(x)$ is the local electrostatic potential measured relative to the bulk and z_i is the ion charge number. The width of this layer is the **Debye length**:

$$\lambda_D = \sqrt{\frac{\varepsilon RT}{F^2 \sum_i z_i^2 c_i^{\mathrm{bulk}}}}, \tag{7}$$

which for our 0.3 M ionic strength electrolyte gives $\lambda_D \approx 0.55$ nm. Submerged in this region, a continuum description still works (it has been re-derived from first principles; see Bard & Faulkner Ch. 13).

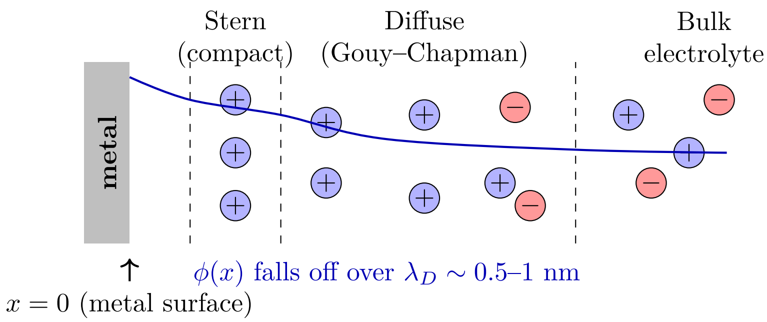
Stern (1924). Combines the two: a rigid **compact layer** (Stern layer) of finite thickness right against the electrode, beyond which the Gouy–Chapman diffuse layer takes over. The Stern layer is treated as a parallel-plate capacitor with capacitance C_S (F/m^2). Most of the applied voltage drops across the Stern layer at high overpotential.

Bikerman (1942). Adds the missing physics that ions have finite size: the exponential Boltzmann factor (6) cannot pile up arbitrary amounts of counter-ion at the surface, because once you hit close-packing the local steric pressure resists further crowding. Bikerman's modification:

$$c_i(x) = \frac{c_i^{\mathrm{bulk}}\, e^{-z_i F \phi(x)/RT}}{1 + \sum_j \frac{c_j^{\mathrm{bulk}}}{c_{\max}} \left(e^{-z_j F \phi(x)/RT} - 1\right)}. \tag{8}$$

At large $|\phi|$, this saturates at $c_i \to c_{\max}$ instead of diverging. The codebase uses a multi-species Bikerman closure defined in `Forward/bv_solver/boltzmann.py`; we will look at it in doc 03.

3.3 A picture of the EDL



The metal sits at $x = 0$ at the left. The Stern layer is one or two ionic radii thick (about 0.3 nm). Beyond it the diffuse layer extends a few Debye lengths into the electrolyte before the potential and concentrations relax to their bulk values.

Exercise 3.1. For $z_i = +1$, $V_T = 25.7$ mV at room temperature, what is the ratio $c_i(0)/c_i^{\text{bulk}}$ from the plain Boltzmann form (6) at $\phi(0) = -0.3$ V? Why is this physically absurd, and what does Bikerman do about it?

4 Electrochemical potential

4.1 Definition

When a charged species i sits in a region of nonzero electric potential ϕ, its true thermodynamic "willingness to be there" is not just its chemical potential μ_i but the **electrochemical potential**:

$$\tilde{\mu}_i = \mu_i^\circ + RT \ln a_i + z_i F \phi. \tag{9}$$

This is the central thermodynamic quantity of the entire field, and it is also the variable our solver uses internally for H^+ (the so-called `logc_muh` formulation: "log-c with μ_H"). We will see in doc 02 why that choice makes Newton's method behave better.

At equilibrium, $\tilde{\mu}_i$ is uniform across the electrolyte. So if you know $\phi(x)$, you can solve for $c_i(x)$ from $\tilde{\mu}_i = \tilde{\mu}_i^{\text{bulk}}$, which immediately recovers the Boltzmann distribution (6).

4.2 The thermal voltage V_T as a unit

Notice that everywhere $zF\phi$ appears, it is divided by RT to form $z\phi/V_T$. So in our nondimensionalized code, voltages are always reported in units of V_T, and the variable `phi_hat` $= \phi/V_T$ is what the residual sees. Doc 02 has the full nondimensionalization.

5 Butler–Volmer kinetics

5.1 The classical form

For a single elementary electron-transfer reaction

$$\mathrm{O} + n_e e^- \rightleftharpoons \mathrm{R},$$

transition-state theory (sketch in any electrochemistry textbook, e.g. Bard & Faulkner Ch. 3) gives the net forward current density:

$$i = i_0 \left[\exp\left(\frac{(1-\alpha)n_e F\eta}{RT}\right) - \exp\left(\frac{-\alpha n_e F\eta}{RT}\right)\right], \tag{10}$$

where:

- $\eta = V - E_{\mathrm{eq}}$ is the overpotential.
- $\alpha \in (0,1)$ is the **symmetry factor** or **transfer coefficient** (often $\alpha \approx 0.5$).
- i_0 is the **exchange current density** at $\eta = 0$; it depends on k_0 (the rate constant) and the local concentrations.

We use a slightly different decomposition that scales better numerically (the "log-rate" form, `bv_log_rate=True`):

$$\log r = \log k_0 + \sum_p p \cdot \log a_p - \alpha n_e \frac{\eta}{V_T}, \tag{11}$$

where p runs over the species in the rate law with stoichiometric coefficient ν_p. The reason for log-rate is that k_0 varies over many orders of magnitude in our calibration, and Newton iterates on $\log r$ stay smooth even in regions where r itself underflows double precision.

5.2 Tafel limits

When $|\eta|$ is large enough that one of the exponentials in (10) dwarfs the other, the equation reduces to the **Tafel form**:

$$\log_{10}|i| \approx \log_{10} i_0 + \frac{(1-\alpha)n_e}{2.303\, V_T}\,\eta \quad \text{(anodic)},$$

giving a straight line of slope $\sim 60/[(1-\alpha)n_e]$ mV/decade. Plotting $\log|i|$ vs. η and fitting straight-line slopes is how experimentalists historically measured α and k_0. Look up **Tafel slope** when you see it in commit messages.

5.3 Multi-step ORR and the parallel topology

The ORR is not a single elementary step. It proceeds through several adsorbed intermediates:

$$O_2 \to {}^*O_2 \to {}^*OOH \to {}^*O \to {}^*OH \to H_2O,$$

where * denotes a surface-adsorbed species. At any point along this chain the system can branch off: if the $*$OOH intermediate desorbs as H_2O_2, you exit early with the 2-electron product; if it stays bound and gets further reduced, you proceed to water through the 4-electron path. Following Ruggiero 2022 §1 we do not resolve the intermediates; instead we lump them into two *parallel* effective Butler–Volmer reactions, $\mathbf{R_{2e}}$ and $\mathbf{R_{4e}}$, each with its own E°, k_0, and α:

$$\mathrm{R_{2e}}: \qquad O_2 + 2H^+ + 2e^- \to H_2O_2 \qquad E^\circ = 0.695\ V_{\mathrm{RHE}} \tag{12}$$

$$\mathrm{R_{4e}}: \qquad O_2 + 4H^+ + 4e^- \to 2\,H_2O \qquad E^\circ = 1.23\ V_{\mathrm{RHE}} \tag{13}$$

Both reactions consume O_2. They share no surface intermediate in the lumped model. The total disk current is $i_D = 2Fr_{2e} + 4Fr_{4e}$. The peroxide that escapes is $2Fr_{2e}$.

Historical hazard: the codebase used to use a different, *sequential* reaction set (R_0: $O_2 \to H_2O_2$; R_1: $H_2O_2 \to H_2O$). That set was wrong — it conflicts with Ruggiero 2022 — and was retired in May 2026 (commit M3a.2). You will still see "R_0/R_1" in old commits, old plot legends, and old constants like `E_eq_r1=0.68`. **All deck-aligned work uses the parallel 2e/4e set.**

5.4 Selectivity and electron number

In an RRDE experiment (next section) you measure two currents: the disk current I_D (negative, all the cathode reduction at the disk) and the ring current I_R (positive, peroxide that escaped is being re-oxidized at the downstream ring). From these:

$$\mathrm{H_2O_2}\,\% = \frac{200 \cdot (I_R/N)}{|I_D| + I_R/N} \tag{14}$$

$$n_e = \frac{4 \cdot |I_D|}{|I_D| + I_R/N} \tag{15}$$

where $N \in (0, 1)$ is the geometry-determined **collection efficiency**: only a fraction of the peroxide made at the disk reaches the ring before being washed downstream. For our deck, $N = 0.224$.

If *all* of the disk current came through R_{2e}, selectivity is 100% and $n_e = 2$. If *all* comes through R_{4e}, selectivity is 0% and $n_e = 4$. Real experiments fall in between.

Exercise 5.1. Derive the factor of 200 in (14) by counting electrons. Hint: you have to correct I_R by $1/N$ to recover the actual H_2O_2 flux out of the disk; then the ratio (peroxide-equivalent current at the disk) / (total disk current as 4e-equivalent) gives the fraction.

6 Mass transport: how reactants reach the electrode

The BV equation tells you the rate at which the surface chemistry *would* consume reactants if there were unlimited supply. In reality, O_2 has to diffuse from the bulk to the electrode and H^+ has to as well. Above some current density, you run out of reactant at the surface and the kinetic equation is no longer the bottleneck — you are **mass-transport limited**.

6.1 The flux equation: Nernst–Planck

For a dilute species i, the flux per unit area is the sum of three contributions:

$$\mathbf{J}_i = \underbrace{-D_i \nabla c_i}_{\text{diffusion}} \underbrace{- \frac{z_i F}{RT} D_i c_i \nabla \phi}_{\text{migration}} + \underbrace{c_i \mathbf{v}}_{\text{convection}} . \tag{16}$$

Mass conservation gives $\partial_t c_i + \nabla \cdot \mathbf{J}_i = 0$. This is the **Nernst–Planck equation**, the "NP" in PNP. We almost always work in steady state ($\partial_t = 0$) and 1D, so this becomes a coupled boundary-value problem.

6.2 The rotating disk electrode (RDE)

The disk in our experiment rotates at $\omega = 1600$ rpm. This creates a well-defined hydrodynamic flow that brings fresh electrolyte from the bulk toward the disk and then flings it radially outward. Levich (1962) showed that the convection and the diffusion combine into an effective **diffusion layer thickness**:

$$\delta \approx 1.612\, D_i^{1/3} \nu^{1/6} \omega^{-1/2}, \tag{17}$$

where ν is the kinematic viscosity of the electrolyte. For O_2 in our system, $\delta \approx 16\ \mu$m at 1600 rpm.

The corresponding maximum (Levich-limited) current density when the surface reactant is fully depleted is:

$$i_{\text{lim}} = 0.62 \cdot n_e F\, D_i^{2/3} \nu^{-1/6} \omega^{1/2}\, c_i^{\text{bulk}}. \tag{18}$$

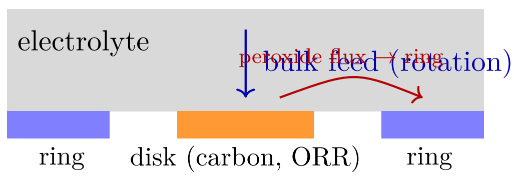
For our O_2 system at 1600 rpm, $i_{\text{lim}}^{(4e)} \approx 5.7\ \text{mA/cm}^2$. Whenever you see a plateau in a current-voltage curve, the first thing to check is whether you have hit Levich.

6.3 Why we use a 1D slab geometry in the model

The RDE setup creates flow that is "**uniformly accessible**" across the disk: every point on the disk surface sees the same δ and the same bulk feed. So we can model the whole disk problem as a 1D slab from $x = 0$ (electrode) to $x = L_{\text{eff}}$ (bulk), with bulk concentrations as Dirichlet boundary conditions at $x = L_{\text{eff}}$. The convection term in (16) is absorbed by choosing L_{eff} to match the Levich δ. Our default is $L_{\text{eff}} \approx 16$–$100\,\mu\text{m}$ depending on the study.

6.4 The ring of the RRDE

Add a second concentric Pt ring electrode just downstream of the disk, held at a much more positive voltage (+1.2 V vs. RHE) where *any* H_2O_2 that reaches it gets oxidized back to O_2. The ring current I_R then directly measures the flux of peroxide leaving the disk. The fraction of disk-released peroxide that reaches the ring is the *collection efficiency* N you saw above (eq. 14). Geometry-only; calibrated with a known redox couple.



Exercise 6.1. If $|I_D| = 2.5$ mA, $I_R = 0.090$ mA, $N = 0.224$: compute selectivity H_2O_2% and effective electron number n_e.

7 Cation effects: the Phase 6β story

This section is the bridge to current research. It is the *specific* chemistry our group is working on, and you will see all of these terms in commits, plots, and the `docs/phase6/` directory.

7.1 The empirical observation

Hold everything fixed — catalyst, pH, total ionic strength, applied voltage — and only change which alkali cation is in the salt: $Li_2SO_4 \to Na_2SO_4 \to K_2SO_4 \to Cs_2SO_4$. The disk current and the peroxide selectivity shift substantially. This is the **cation effect**. It has been reproducibly observed for decades and across many catalysts. The mystery: cations don't appear in the ORR half-reaction.

7.2 The currently leading hypothesis: cation hydrolysis at the OHP

The hypothesis our group works under (following Singh 2016, Co & Zhang 2019, Ringe 2019) is that the "inert" cation participates in a side reaction at the outer edge of the Stern layer (the **outer Helmholtz plane**, OHP). The hydrated cation $\mathrm{M(H_2O)_n^+}$ can give up a proton in a hydrolysis equilibrium:

$$\mathrm{M(H_2O)_n^+ + H_2O \;\rightleftharpoons\; M(OH)(H_2O)_{n-1}^0 + H_3O^+}. \tag{19}$$

Two things make this matter near a polarized cathode:

1. **Field-dependent pK$_a$.** Singh 2016 derived that the bulk pK$_a$ of (19) shifts by an electric-field-dependent amount near the cathode:

$$\mathrm{pKa}(\sigma) = \mathrm{pKa}_{\mathrm{bulk}} + 2Az\sigma\, r_{H\text{–}El}\left(1 - \frac{r^2_{M-O}}{r^2_{H\text{–}El}}\right), \tag{20}$$

with σ the surface charge density at the electrode, r_{M-O} the cation–O distance, $r_{H\text{–}El}$ the hydration-shell-H-to-electrode distance, and $A = 620.32$ pm. The geometry makes $\Delta\mathrm{pKa} < 0$ for cathodic $\sigma > 0$: the hydrolysis becomes much more favorable near the surface than in the bulk. For Cs^+ near a Cu cathode, pK$_a$ drops from 14.8 in bulk to ~ 4.3 at typical ORR field strengths.

2. **Cation identity sets the bulk pK$_a$.** Smaller, more strongly hydrating cations (Li^+) have lower bulk pK$_a$ values; larger ones (Cs^+) have higher ones (see Singh Table S1). At a fixed bulk pH and fixed σ, this changes *which* cations are actively releasing protons at the OHP and which are not.

The protons released by (19) are released exactly where the ORR cathodic chemistry is consuming them, locally shielding the surface from the otherwise severe alkalinization ($H_2O + e^- \to \frac{1}{2}O_2 + OH^-$ raises local pH a lot). This converts the cation from a passive spectator into an active, local pH-buffering species, even though it never enters the formal ORR equation.

7.3 Why this is hard to model

The cation enters through:

- the EDL physics (Bikerman steric crowding at the OHP),
- a coupled **coverage** variable Γ_{MOH} on the OHP that obeys its own ODE in time (rate of (19) forward vs. reverse, both functions of the local field, which is itself a function of Γ via charge balance, etc.),
- a feedback into the proton residual at the boundary that changes the BV reaction rates.

This nonlinear feedback loop is what **Phase 6β v9**, **v10a**, and **v10b** have been progressively wiring into the solver. Doc 04 walks the chronology; for now, the bullet to remember is: *the next big science our model needs to reproduce is cation-dependent peroxide selectivity, and the mechanism we are betting on is field-dependent hydrolysis of hydrated alkali cations at the OHP.*

Exercise 7.1. Given Singh's bulk pK$_a$ values {Li: 13.6, Na: 14.2, K: 14.5, Cs: 14.8} and a uniform $\Delta\mathrm{pKa} = -10$ shift near a polarized Cu cathode, which of the four cations is most strongly buffering at $V_{\mathrm{RHE}} = -0.2$ V in a pH 4 bulk? (Hint: ask yourself which cations have a local pK$_a$ near or below the local pH, since only those will donate appreciable proton.)

8 Glossary of terms you will see in commits

Term	Meaning
ORR	Oxygen reduction reaction. $O_2 \rightarrow H_2O$ or H_2O_2.
RRDE	Rotating ring–disk electrode — the experimental setup.
V_{RHE}	Voltage vs. reversible hydrogen electrode.
η	Overpotential, $V - E_{\mathrm{eq}}$. Negative is cathodic.
k_0	Reaction-rate prefactor (intrinsic activity).
α	Symmetry factor in BV, typically ~ 0.5.
E°	Standard reduction potential.
EDL	Electric double layer.
OHP	Outer Helmholtz plane: the boundary between the Stern and diffuse layers.
C_S	Stern (compact-layer) capacitance, F/m^2. We use 0.20 F/m^2.
λ_D	Debye length: width of the diffuse layer, ~ 0.55 nm here.
δ	Levich diffusion-layer thickness, ~ 16 μm at 1600 rpm.
L_{eff}	Slab thickness in our 1D model, set to δ.
N	RRDE collection efficiency, 0.224 here.
n_e	Effective electron number, between 2 and 4.
PNP	Poisson–Nernst–Planck. The PDE system.
BV	Butler–Volmer. The kinetic boundary condition.
Bikerman	Steric correction to the Boltzmann distribution.
Phase 6α	Project epoch: water self-ionization landed.
Phase 6β	Project epoch: cation hydrolysis at OHP (*ongoing*).

9 Where to read more

- Bard & Faulkner, *Electrochemical Methods* (2nd ed.): Chapters 1, 3, 12, 13. The standard textbook; chapters 1 and 3 give you everything in this primer in proper depth.
- Ruggiero 2022 PDF (`docs/papers/Ruggiero2022_JCatal_manuscript.pdf`): our experimental anchor. Read §1, §2, and Fig. 1B.
- Internal cheat-sheet (`docs/papers/Ruggiero2022_JCatal_source_paper.md`): compact list of every number in the paper we use.
- Singh 2016 SI extraction (`docs/phase6/singh_2016_pka_formula.md`): how Eq. 20 is derived and verified against experimental titration data for the alkali metals.
- CMK-3 capacitance literature (`docs/phase6/CMK3_capacitance_literature.md`): the citation chain behind our $C_S = 0.20$ F/m^2 choice.